FOLLOW THE REGULARIZED LEADER

- To overcome the shortcomings of the FTL algorithm, one can incorporate a regularization function $\psi : \mathcal{A} \to \mathbb{R}_+$ into the action choice of FTL, which leads to more stability.
- To be more precise, let for $t \geq 1$

$$a_t^{\mathrm{FTRL}} \in \operatorname*{arg\,min}_{a \in \mathcal{A}} \left(\psi(a) + \sum_{s=1}^{t-1} (a, z_s) \right),$$

(Technical side note: if there are more than one minimum, then one of them is chosen.)

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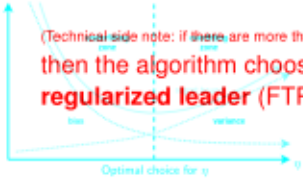

Advanced Machine Learning

Follow the regularized leader

Learning goals

- Get to know FTRL as a useful alternative for FTL
- See a suitable regularization for OLO problems

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REGULARIZATION IN ONLINE LEARNING VS. BATCH LEARNING

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REGRET ANALYSIS OF FTRL: A HELPFUL LEMMA

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(The proof is similar.)

FTRL FOR ONLINE LINEAR OPTIMIZATION

- In the following, we analyze the FTRL algorithm for the linear loss $\langle a, z \rangle = a^\top z$ for online linear optimization (OLO) problems.
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- **Proposition:** Using the FTRL algorithm with the squared L2-norm regularization on any online linear optimization (OLO) problem with $\mathcal{A} \subseteq \mathbb{R}^d$ leads to a regret of FTRL with respect to any action $\tilde{a} \in \mathcal{A}$ of

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- We will show the result only for the case $\mathcal{A} = \mathbb{R}^d$.
- For the more general case, where $\mathcal{A}$ is a strict subset of $\mathbb{R}^d$, we need a slight modification of the update formula above:

$$a_t^{\mathrm{FTRL}} = \Pi_{\mathcal{A}}\big(-\eta \sum\nolimits_{i=1}^{t-1} z_i\big) = \arg\min_{a \in \mathcal{A}} \left\| a - \eta \sum\nolimits_{i=1}^{t-1} z_i \right\|_2^2 .$$

In words, the action of the FTRL algorithm has to be projected onto the set $\mathcal{A}$. Here, $\Pi_{\mathcal{A}} : \mathbb{R}^d \to \mathcal{A}$ is the projection onto $\mathcal{A}$.

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  Reminder (1): $R_T^{\mathrm{FTRL}}(\tilde{a}) \leq \psi(\tilde{a}) - \psi(a_1^{\mathrm{FTRL}}) + \sum_{t=1}^{T} \left((a_t^{\mathrm{FTRL}}, z_t) - (a_{t+1}^{\mathrm{FTRL}}, z_t) \right).$

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- **Proof:**

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FTRL FOR OLO: THEORETICAL GUARANTEES

- Interpretation of the terms in the proposition, i.e., of

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- Interpretation of the impact in the regret bound, i.e., of $\frac{1}{2\eta}||\tilde{a}||_2^2 + \eta$
  Under certain assumptions we can balance the trade-off induced by the bias and the variance by choosing η appropriately.

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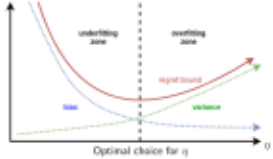

- **Corollary:** Suppose we use the FTRL algorithm with the squared L2-norm regularization on an online linear optimization problem with $\mathcal{A} \subset \mathbb{R}^d$ such that
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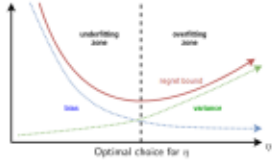

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  Then, by choosing the step size η for FTRL as $\eta = \frac{B}{V\sqrt{2\,T}}$ it holds that

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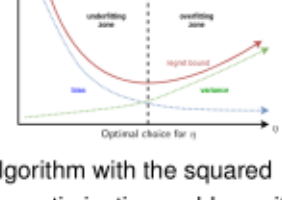

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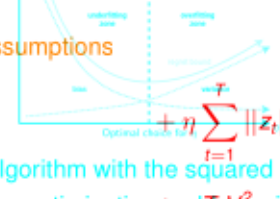
- Note that the (optimal) parameter η depends on the time horizon T, which is oftentimes not known in advance. However, there are some tricks (i.e., the *doubling trick*), which can help in such cases.

FTRL FOR OLO: THEORETICAL GUARANTEES

- **Proof:** Under certain assumptions we can balance the trade-off induced by the bias and the variance by choosing η appropriately.
  - By the latter proposition and the assumptions



$$R_T^{FTRL}(\tilde{a}) \leq \frac{1}{2\eta}\|\tilde{a}\|_2^2 + \eta \sum_{t=1}^{T} \|z_t\|_2^2$$

$$\leq \frac{B^2}{2\eta} + \eta T V^2.$$

- **Corollary:** Suppose we use the FTRL algorithm with the squared L2-norm regularization on an online linear optimization problem with $\mathcal{A} \subset \mathbb{R}^d$ such that
  - $\sup_{\tilde{a} \in \mathcal{A}} \|\tilde{a}\|_2 \leq B$ for some finite constant $B > 0$,
  - $\sup_{z \in \mathcal{Z}} \|z\|_2 \leq V$ for some finite constant $V > 0$.

  Then, by choosing the step size η for FTRL as $\eta = \frac{B}{V\sqrt{2T}}$ it holds that

$$R_T^{FTRL} \leq BV\sqrt{2T}.$$

- Note that the (optimal) parameter η depends on the time horizon T, which is oftentimes not known in advance. However, there are some tricks (i.e., the *doubling trick*), which can help in such cases.

FTRL FOR OLO: THEORETICAL GUARANTEES

- **Proof:**
  - By the latter proposition and the assumptions

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&\leq \frac{B^2}{2\eta} && + \eta\, T\, V^2.
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DESIRED RESULTS THEORETICAL GUARANTEES

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